MIL-A-27261A(USAF)
8 November 1966
Superseding
MIL-A-27261 (USAF)
15 October 1959

MILITARY SPECIFICATION

ACCELEROMETER, AIRCRAFT

1. SCOPE

*1.1 This specification covers integrally lighted, 1-7/8-inch dial aircraft accelerometers having a range of -2 to +4g's.

*1.2 Classification. Accelerometers shall be of the following types, as specified (see 6.2):

ABU-8/A	Cover glass coated with antireflection coating per MIL-C-675
ABU-8A/A	Cover glass coated with multilayer antireflection coating of high efficiency (superior to coating specified for ABU-8/A accelerometer).

* 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Codmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection)
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Orders of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-781	Reliability Tests Exponential Distribution

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

* 3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.3.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the accelerometer except where ferrous materials are essential.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.6 Corrosive fumes. The materials, as installed in the accelerometer and under the service conditions specified herein, shall not liberate deleterious fumes.

3.3.7 Protective treatment. When materials are used in the construction of the accelerometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction

3.4.1 Design. The accelerometer shall conform to figure 1 and shall be designed

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

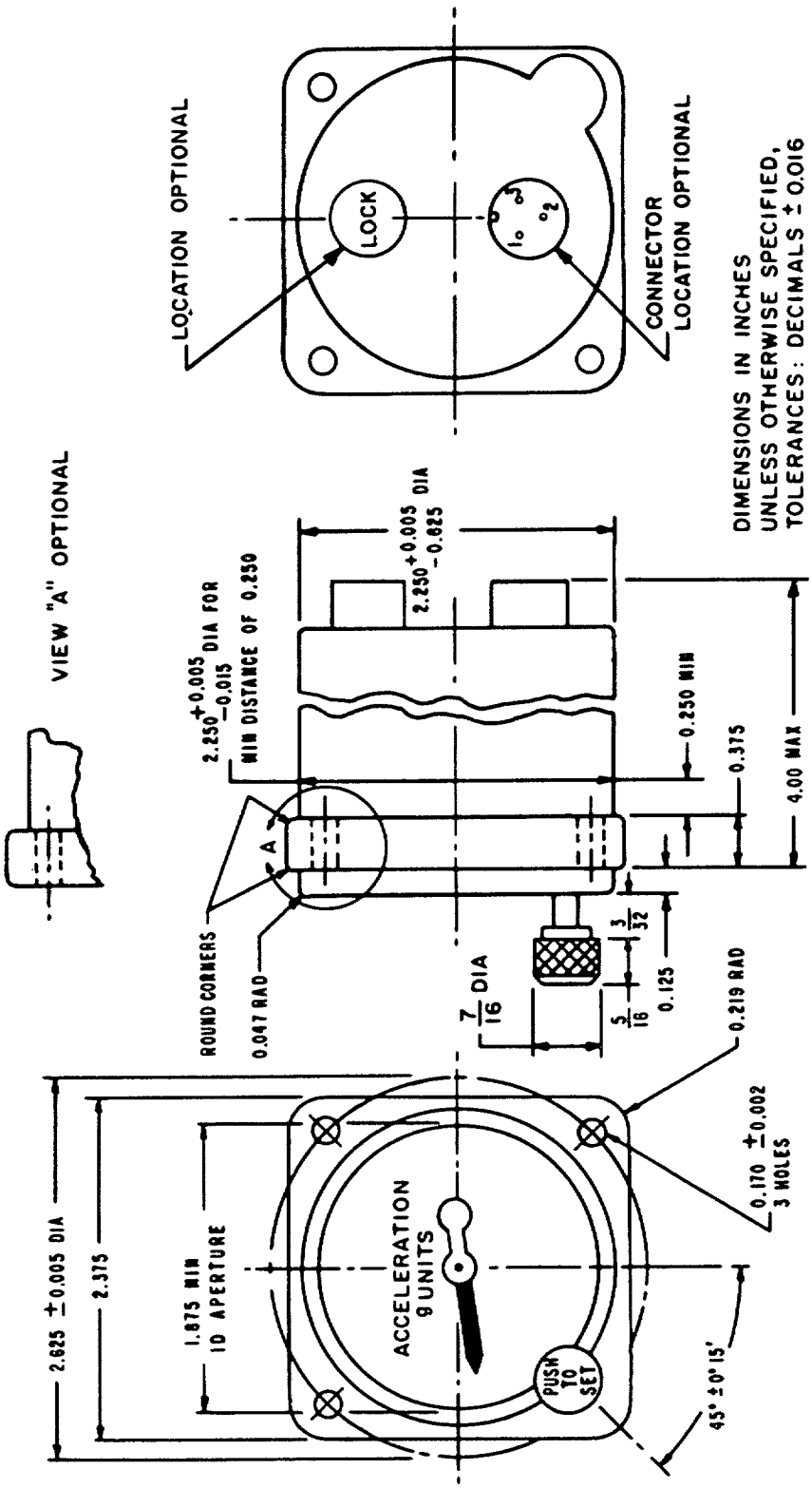


Figure 1. Case

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

to indicate accelerations during maneuvers and in rough air over a range of -2 to +4gs. It shall have a natural frequency of 6 to 8 cps and 0.5 to 0.7 critical damping, and shall indicate vertical accelerations when the plane of the mounting lugs is vertical and the +1 graduation is 270°. In this position, normal gravity shall be indicated at +1g.

* 3.4.1.1 Reliability. The accolerometer shall have a mean-time-between-failures (MTBF) of 400 hours at a discrimination ratio of 1.5.

3.4.2 Construction. The accelerometer shall be of simple construction to permit overhaul and repair without the use of special tools and fixtures. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. Pivots, bearings, and gears shall neither bind nor shake and shall be as nearly frictionless as practicable.

* 3.5 Performance. The accelerometer shall be capable of operating as specified herein under the following conditions:

a. Operating temperatures ranging from -54° to +71°C and nonoperating temperatures ranging from -62° to +71°C.

b. Relative humidity up to 100 percent

c. Pressures ranging from 30 inches Hg down to 1.31 inches Hg (approximately 70,000 feet)

d. Fungus growth as encountered in tropical climates

e. Alternate immersions in water at 85°C and 5°C for eight cycles

f. Exposure to salt sea atmosphere for a period of 50 hours

g. Shock forces of 20g's

h. Salt fog for a period of 24 hours.

* 3.5.1 Vibration. When vibrated at an amplitude of 0.010 to 0.020 inch at frequencies varied between 500 and 3,000 cpm for a total of 3 hours, the main pointer shall not oscillate more than ±0.3g. After vibration, the auxaliary pointers shall not have changed more than 0.1g.

3.5.2 Static scale error and friction. Static scale error shall not exceed ±0.1g and friction error shall not exceed 0.2g and 0.3g under the conditions specified in 4.6.2.

3.5.3 Dynamic scale error. Dynamic scale error shall not exceed ±0.2g at any specified test point and shall not exceed ±0.1g at two-thirds of the test points.

3.5.4 Magnetic effect. When the accelerometer is revolved about a magnetic compass with a magnetic field intensity of 0.17 to 0.19 oersted, the compass shall not deflect more than 1°.

3.5.5 Mounting lugs. The mounting lugs shall withstand a 175-pound load without fracture.

3.5.6 Response. When the accelerometer is vibrated at frequencies from 4 to 20 cps at maximum g-forces at each test point, response shall be within the limits specified in 4.6.7.

* 3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

3.7 Case. The case shall conform to figure 1. It shall be hermetically sealed and of such design that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic seal shall be so accomplished that seal will not be dependent upon materials that would be affected by the action of any atmosphere to which the accelerometer might be subjected.

* 3.7.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressures of the filling medium in the case shall be 12 to 14.7 psi.

* 3.7.2 Finish. The case shall be finished in durable dull black, color No. 37038 of FED-STD-595.

* 3.8 Reflectance. The ABU-8/A cover glass shall meet the requirements of MIL-C-675. The ABU-8A/A cover glass shall meet the requirements of MIL-C-675, in addition to withstanding the environmental conditions specified herein, except it shall be coated with a multilayer coating that will meet the following reflectance tolerances:

Angle of Incidence	Wavelength Millimicrons	Percent Reflectance
0°	450-675	0.6 maximum
0°	425-700	0.5 average
30°	450-625	1.0 maximum
30°	425-700	0.5 average.

3.9 Dial. The dial shall conform to figure 2. It shall be fastened securely to the case or on the mechanism frame by not less than two screws in such a manner that it will not loosen or turn when the instrument is vibrated.

3.9.1 Distance from dial to cover glass. The distance between the dial and cover glass shall be not greater than 0.250 inch.

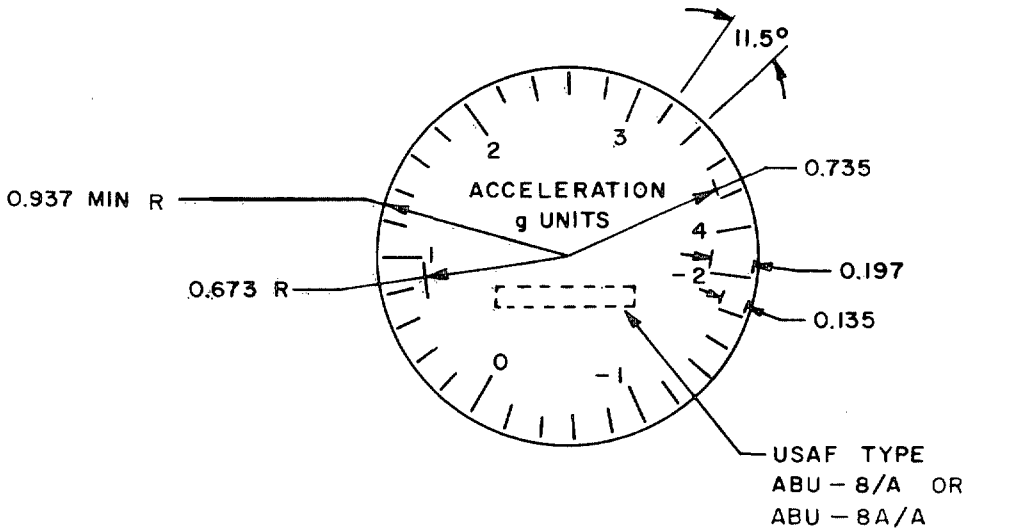
3.9.2 Numerals and letters. The style and proportion of numerals and letters, except the letter g, placed on the dial of the accelerometer shall conform to MS33558. Each numeral shall distinctly indicate the graduation to which it applies.

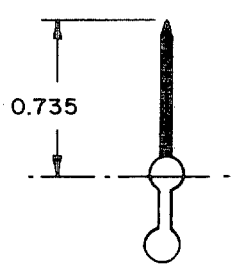
3.10 Pointers. The accelerometer shall have three pointers; one main pointer for indicating instantaneous acceleration and two auxiliary pointers; one for indicating maximum positive acceleration; and one for indicating maximum negative acceleration. The pointers shall conform to figure 2, and shall be light and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the mechanism, yet shall be readily adjustable.

3.10.1 Ratchet mechanism. Each auxiliary pointer shall be provided with a ratchet device so that indication of maximum plus and minus deflections of the main pointer will be maintained until the auxiliary pointers are reset to obtain a new reading.

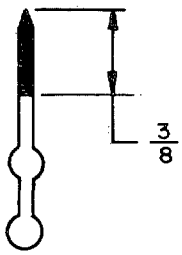
3.10.2 Auxiliary pointer return mechanism. A mechanism shall be provided for resetting the auxiliary pointers into juxtaposition with the main pointer. A knob, located as shown on figure 1, shall be provided for operating the return mechanism. Stops shall be provided so that when the auxiliary pointers are reset, they will not indicate a lower or higher reading than that indicated by the main pointer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.





MAIN POINTER
MS33585-1



AUXILIARY POINTER
MS33585-1
EXCEPT AS SHOWN

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED,
TOLERANCES:

FRACTIONS $\pm \frac{1}{64}$

DECIMALS $\pm$ 0.010

ANGLES $\pm$ 0° 10'

Figure 2. Dial and Pointers

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

Stops shall be a part of the mechanism connecting the pointers and shall not directly engage them. Operation of the return mechanism shall require 2-1/2 to 8 pounds of force on the knob.

3.10.3 Main pointer mechanism. The main pointer mechanism shall not engage the mechanism of the auxiliary pointers when the imposed acceleration is less than the acceleration indicated by the auxiliary pointers.

3.11 Acceleration-responsive element

3.11.1 Stops. Mechanical stops shall be provided for arresting the motion of the acceleration-responsive element when the accelerometer is subjected to accelerations beyond the limits specified. One stop shall engage the element at an acceleration of not less than -2.1g and not more than -2.2g, and the other stop shall engage the element at an acceleration of not less than +4.1g and not more than +4.2g.

3.11.2 Locking device. A device for manually locking the position of the acceleration-responsive element shall be provided to protect the mechanism of the accelerometer from possible damage during shipment. This device shall be controlled by means of a knurled knob located in the back of the case.

3.11.3 Adjustment. The mechanism shall be provided with means to adjust or correct the indications. The means of adjustment shall be reasonably simple to manipulate and of such a nature that adjustment can be accomplished with tools ordinarily used by an instrument repairman.

3.12 Marking. All markings shall be durable to withstand the usage encountered in service. Numerals shall distinctly indicate the graduation to which each applies.

* 3.12.1 Color and dimensions. The following markings shall be finished in dull white, color No. 37875 of FED-STD-595. The marking dimensions shall be as specified:

	Height or Length Inch ±1/64	Width of Line or Graduation Inch ±0.005
Numerals	3/16	MS33558
Graduations corresponding to numerals	See figure 2	0.031
Lettering g UNITS and ACCELERATION	1/16	MS33558
One g graduation	---	0.062
Shaded part of pointers	---	---
Knob markings	---	---
Small graduations	See figure 2	0.015

* 3.12.2 Other markings. The following markings shall be durably and legibly marked on the dial. These markings, the background of the dial, and all other markings not otherwise specified shall be finished in durable dull black, color No. 37038 of FED-STD-595.

	Height or Length Inch - 1/64
Lettering USAF and TYPE ABU-8/A or ABU-8A/A, as applicable	1/16

* 3.12.3 Visibility of dial. The pointers, numberals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone the side of which makes an angle

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

of 20° with a perpendicular to the dial and of which the small diameter is the aperture of the instrument case.

3.13 Intergral lighting. The accelerometer shall be integrally lighted with 2 parallel lighting circuits, one white and one red as shown on figure 3. Red lighting shall be in accordance with MIL-L-25467 and white lighting shall be in accordance with MIL-L-27160. In the event of conflict between this specification and either MIL-L-25467 or MIL-L-27160, this specification shall govern. Although resistors will not be permitted in the original design of the lighting systems, space shall be allocated for incorporation of a trimming resistor in each lighting circuit within the indicator. The magnitude of the trimming resistors shall be specified by the procuring activity in terms of the brightness decrease required. The lighting shall be subject to approval by the procuring activity.

3.13.1 Prime standard. The integral lighting shall be comparable to and shall balance in brightness with a prime standard which shall be an accelerometer of the type specified herein. The specific voltage which the procuring activity has established as necessary to assure equality of presentation with respect to the lighting of adjacent but dissimilar instruments shall be applied to the prime standard. The contractor shall submit the test procedure for determining compliance with this requirement to the procuring activity for approval.

* 3.14 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454.

* 3.15 Lubrication. The accelerometer shall be lubricated in accordance with MIL-STD-838.

3.16 Weight. The weight of the accelerometer shall not exceed 1.25 pounds.

3.17 Finishes and protective coatings

3.17.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

3.17.1.1 Dials, small holes, and case inserts need not be anodized.

3.17.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.17.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.17.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.17.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.17.2 Steel parts. Where practicable, steel parts shall be cadmium plated in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

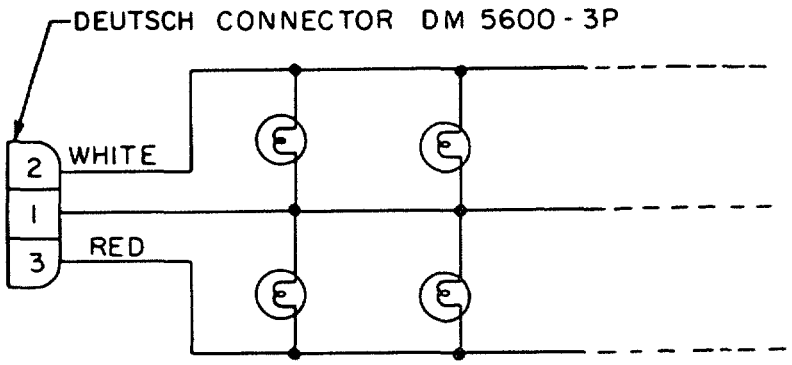


Figure 3. Lighting Circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

3.19 Workmanship. The accelerometer, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, and riveting; alignment of parts and tightness of assembly screws and bolts.

3.19.1 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.19.2 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.19.3 Gears. Gear assembles shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free from backlash.

3.19.4 Cleaning. The accelerometer shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign material removed after final assembly. Burrs and sharp edges as well as resin flash that may crumble shall also be removed.

3.19.5 Dimensions. Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the accelerometer, they shall be held or limited accordingly.

3.20 Envelope. An envelope containing the mounting screws shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of accelerometers shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

4.3.2 Operating position and tapping. Unless otherwise specified, the accelerometer shall be tested in the normal operating position and shall be gently tapped or vibrated before a test reading is taken. Tapping (see 6.4a) shall be applied in a longitudinal direction with respect to the dial of the instrument.

4.3.3 Vibration stand. The vibration stand used shall vibrate at any desired frequency between 500 and 3,000 cpm and shall subject the accelerometer to vibration in such a manner that a point on the case will describe a circle, the diameter of which shall be equal to the amplitude specified herein, in a plane inclined 45° to the horizontal plane.

4.3.4 Calibration. Calibration shall be determined at all test points from -2g to +4g in increments of 1g.

4.3.5 Response characteristics test apparatus. The test apparatus shall impose linear sinusoidal motion to the accelerometer and shall produce the frequency and double amplitude required for each test point. The acceleration vector of any harmonic measured on the test table shall not exceed 10 percent of the fundamental acceleration vector.

4.3.6 Adjustments and repairs. No adjustments or repairs of any kind shall be performed on the test samples after starting preproduction testing.

4.4 Preproduction testing

* 4.4.1 Test samples. The test samples shall consist of three accelerometers representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. The contractor shall submit the samples that were tested to the procuring activity in the "as is" condition.

* 4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.3).

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each accelerometer shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Static scale error and friction

c. Dynamic scale error

d. Sealing

e. Return mechanism operation

f. Integnal lighting

g. Response characteristics (4.6.7).

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. One accelerometer shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Response characteristics (4.6.8)

c. Low temperature operation

d. High temperature operation

e. Magnetic effect

f. Vibration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

★ 4.5.2.2 Sampling plan B. Unless otherwise specified, 3 accelerometers shall be selected at random from the first 15 on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Shock

c. Mounting lugs

d. Temperature-altitude

e. Humidity

f. Fungus

g. Thermal shock

h. Salt fog.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

★ 4.5.2.4 Sampling plan C. Unless otherwise specified, three accelerometers shall be selected at random from the first 15 produced on the contract or order and an additional 3 accelerometers shall be selected at random from each 100 produced and subjected to the test specified in 4.6.21. Acceptance or rejection shall be as specified therein.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The accelerometer shall be examined externally to determine conformance with this specification with respect to material, workmanship, marking, and envelope.

4.6.2 Static scale error and friction. The position error shall be observed with the accelerometer held in each of the following positions:

a. Plane of dial vertical with +1g graduation at 270°

b. Plane of dial horizontal

c. Plane of dial vertical with +1g graduation at 90°.

In each position, the main-pointer reading obtained shall be due entirely to gravity and not due to a change in the position of the instrument. The static scale error shall not exceed ±0.1g in any of the three position. The accelerometer shall also be tested for friction at the three specified positions. The amount of tapping or vibration applied shall be sufficient to remove all friction from the accelerometer. In going from one position to the next, one of the auxiliary pointers shall be engaged. After the desired test position is reached, two readings shall be taken, the first before vibration and the second after vibration. The difference in the two readings shall be recorded as friction and shall not exceed 0.2g with the plane of the dial vertical nor 0.3g with the plane of the dial horizontal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

4.6.3 Dynamic scale error. The accelerometer shall be mounted on a suitable centrifuge and the scale error determined for accelerations of -2, -1, +1, +2, +3, and +4 and at any other points deemed necessary to determine accuracy. The dynamic scale error shall not exceed ±0.2g at any specified test points and shall not exceed ±0.1g at two-thirds of the test points.

*4.6.4 Sealing. The accelerometer case shall be tested for leaks by means of a mass spectrometer type of helium leak detector. The detected leak rate shall not exceed 0.10 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi.

4.6.5 Return mechanism operation. The accelerometer shall be placed in operating position with the pointers deflected to the maximum indication, the reset knob shall then be operated to return the pointers to juxtaposition with the main pointer. The force applied to the reset knob shall be between 2-1/2 and 8 pounds.

4.6.6 Integral lighting. Each lighting circuit shall be properly connected individually and tested to show compliance with 3.13.

4.6.7 Response characteristics (individual test). The accelerometer shall be subjected to vibration along an axis which is parallel to the motion of the sensing element with the plane of the dial vertical and with the 1g graduation in the 9 o'clock position. The test apparatus used shall be as specified in 4.3.5. The accelerometer shall be tested at frequencies of 4, 6, 8, 12, 16, and 20 cps. The amplitude of vibration at each point shall be such that the maximum g forces practicable for that test point will be obtained. The indicated g shall be recorded and a plot made for each test point of indicated g/ applied g versus frequency. The points so plotted shall fall within the shaded area of the graph on figure 4.

4.6.8 Response characteristics (sampling plan A). This test shall be identical to the test specified in 4.6.5 except that the accelerometer shall be tested at each of a minimum of 17 different frequencies. These tests shall be approximately equally spaced through the frequency range from 4 to 20 cps.

*4.6.9 Low temperature operation. The accelerometer shall be subjected to a low temperature test in accordance with MIL-STD-810, method 502, except that exposure period shall be 8 hours in lieu of 48 hours. Upon completion of this test and while at the specified temperature, the accelerometer shall be subjected to and shall meet the individual tests.

*4.6.10 High temperature operation. The accelerometer shall be subjected to a high temperature test in accordance with MIL-STD-810, method 501 for a period of 8 hours. Upon completion of this test and while at the specified temperature, the accelerometer shall be subjected to and shall meet the individual tests.

4.6.11 Magnetic effect. The accelerometer shall be revolved about a short bar magnetic compass with the nearest part of the indicator 5-1/2 inches from the magnet and in a vertical plane which is a perpendicular disector of the North-South axis of the magnet. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the accelerometer shall be rotated on its horizontal axis until it is in its normal operating position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 1°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

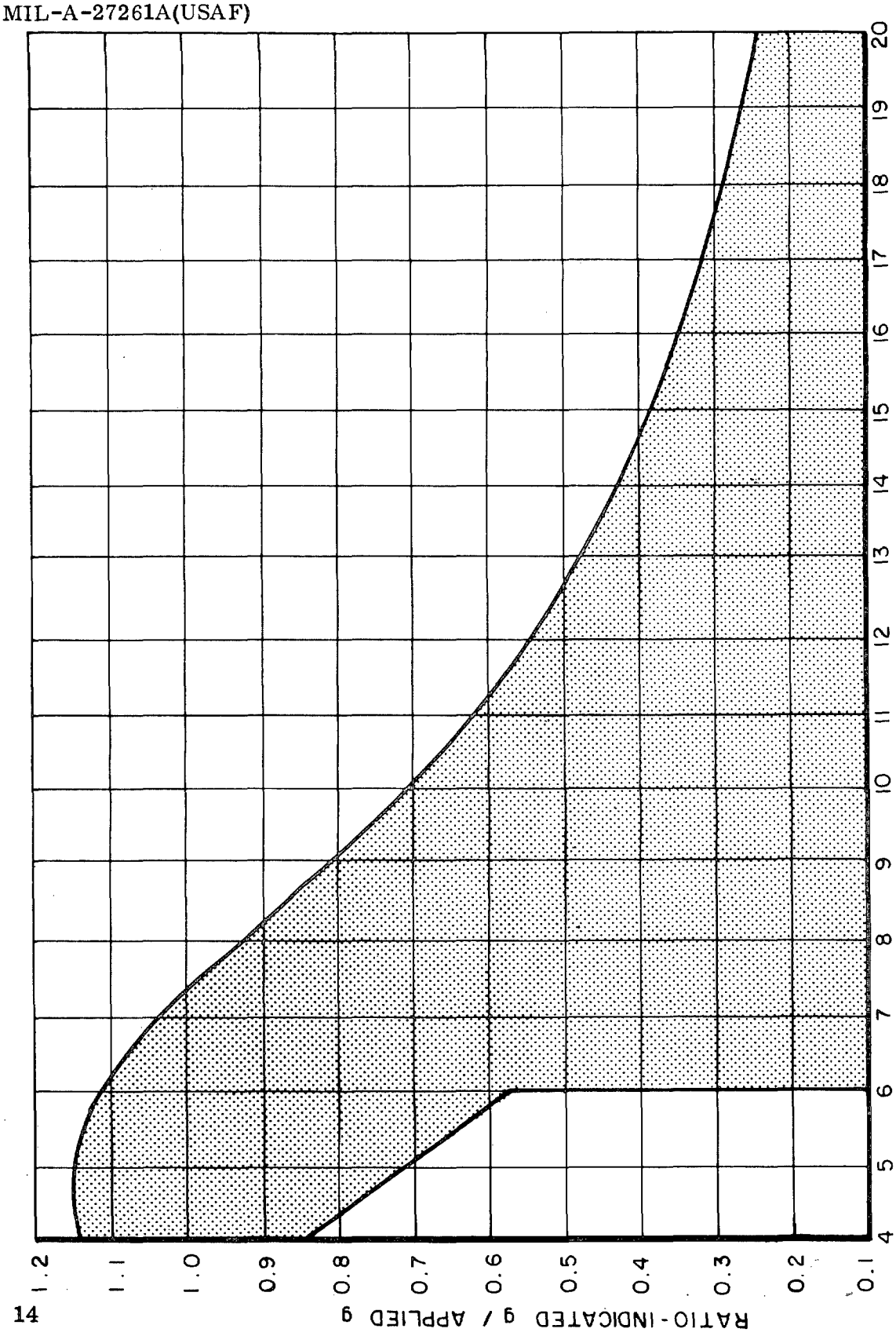


Figure 4. Dynamic Response Characteristics

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

4.6.12 Vibration. The accelerometer shall be mounted on a suitable vibration stand with the dial in a vertical plane and with the graduation +1 at 270°. The auxiliary pointers shall be displaced approximately 1g from the main pointer and the readings noted. The reset mechanism shall not be operated until completion of the tests. For a total of 3 hours, the accelerometer shall be vibrated at the specified frequencies at any amplitude between 0.018 and 0.020 inch, and the pointer oscillation shall be noted. The frequency shall be varied from 500 to 3,000 cpm for a period of not more than 60 minutes during which the natural frequency of the seismic system shall be noted. For the remainder of the 3-hour test period, the accelerometer shall be vibrated at a frequency of 1,800 cpm. At the end of the test period, the indication of the auxiliary pointer shall be read before operating the reset mechanism. At frequencies below 1,200 cpm, the oscillation of the main pointer shall not exceed ±0.2g and at frequencies over 1,200 cpm, the oscillation shall not exceed ±0.3g. After vibration, the indications of the auxiliary pointers shall not have changed by more than 0.1g. The accelerometer shall not be damaged as a result of this test.

*4.6.13 Shock. The accelerometer shall be subjected to a shock test in accordance with MIL-STD-810, method 516, procedure I. The accelerometer mechanism shall be locked during this test.

4.6.14 Mounting lugs. The accelerometer case, with mechanism and cover glass removed, shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs will receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A 175-pound load shall be applied along the axis of the mounting hole and toward the front of the case for a period of 1 minute. There shall be no fracture of the mounting lugs.

*4.6.15 Temperature-altitude. The accelerometer shall be subjected to a temperature-altitude test in accordance with MIL-STD-810 for class 2 equipment, method 504, procedure I, except only steps 1, 2, 3, and 12 shall be conducted. The examination of product and scale error and friction tests shall be conducted after each step while still at the specified ambient condition.

*4.6.16 Humidity. The accelerometer shall be subjected to a humidity test in accordance with MIL-STD-810, method 507, except the number of cyles shall be 4 in lieu of 10.

*4.6.17 Fungus. The accelerometer shall be subjected to a fungus test in accordance with MIL-STD-810, method 508, procedure I.

4.6.18 Thermal shock. The accelerometer shall be subjected to alternate immersions in tap water maintained at 85° ±2°C and 5° ±2°C for a total of eight cycles. The length of time for each immersion shall be 30 minutes with not more than 5 seconds between immersions. There shall be no evidence of moisture penetration or damage to the inclosure. At the conclusion of this test, the accelerometer shall be subjected to and shall meet the test specified in 4.6.4.

4.6.19 Salt fog. The accelerometer shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, for a period of 24 hours. There shall be no evidence of external corrosion or discoloration as a result of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

4.6.20 Reflectance (ABU-8A/A). The reflectance of all glass surfaces of the ABU-8A/A accelerometer shall be measured and the reflectance tolerances shall not exceed those specified in 3.8.

*4.6.21 Reliability. The accelerometers shall be subjected to a reliability test to demonstrate a specified MTBF of 400 hours. The test shall be conducted in accordance with MIL-STD-781, test level E, test plan I, except as modified herein. Each of the three sample accelerometers shall be tested a minimum of 1,240 hours for a minimum total of 3,720 hours. The temperature cycle shall be in accordance with the paragraph entitled "Alternative Method of Temperature Cycling" and the figure specified therein, except the cycling shall be for 10-hour periods. The vibration cycle shall consist of varying the frequency from 1 to 60 cps for a period of 45 minutes of each hour of testing. The applied frequency shall be so varied that frequencies from 1 through 15 cps will be applied for 30 minutes of each hour of testing. Acceleration rates in the 1 to 15 cps range shall be +3.5 -1.5 ±0.5g's. For the vibration portion of this test, the accelerometers shall be mounted in a bracket to simulate panel mounting. Loadings shall be applied along the sensitive axis of the accelerometer in such a manner that the accelerometer will respond to the maximum input loading. During the last 15 minutes of the vibration cycle, the frequency shall be varied from 15 to 60 cps at a constant single amplitude of 0.0095 ±0.0005 inch. The on-off cycle shall be applicable to the lighting circuit only and shall consist of 50 minutes of power on during each hour of testing. After each 100 hours of test time, the accelerometers shall be checked for static and dynamic scale error. After each 400 hours of testing, the accelerometers shall be subjected to the individual tests. In the event a reject decision is reached, i.e., 17 relevant failures or more during 3,720 hours of testing, the units under test shall be modified to eliminate the causes of failure and the test rerun or continued until an accept decision is reached in accordance with the procedures specified herein. In the event a continue-test decision is reached at the end of 3,720 hours of testing, the contractor shall submit all test data to the procuring activity for analysis and continue the test until an accept decision in accordance with test plan I is reached or until 5,360 test hours have been accumulated. If 16 relevant failures or less have been accumulated, the equipment shall be accepted. If 17 or more failures have occurred, the equipment shall be rejected, the equipment under test redesigned or modified to eliminate the cause of failure, and the test rerun until an accept decision is reached in accordance with the procedures specified herein. The contractor may take corrective action on the equipment during any portion of the test provided the procuring activity analyzes and approves such action prior to its incorporation. Subsequent to any corrective action, all equipment already produced shall be modified (see 6.2.1). If 6 or less relevant failures occur during the 3,720 hours of testing, or if 16 or less failures occur during the 5,360 hours of testing, the first 100 units of the contract shall be accepted from a reliability standpoint. In the event the contractor considers a failure nonrevelant, he may submit a failure analysis verifying this conclusion to the procuring activity for approval. If approved, the failure shall be considered nonrelevant.

*4.7 Inspection of preservation, packaging, packing and marking. The inspection of preservation, packaging, packing, and marking for shipment and storage shall be in accordance with section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

*5.1 Preservation, packaging, and packing. Preservation shall be in accordance with MIL-P-116, method II, and packaging and packing shall be in accordance with MIL-STD-794. Levels of packaging and packing shall be as specified by the procuring activity.

*5.2 Marking of shipments. Interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129. The nomenclature shall be as follows, as applicable:

Accelerometer, Aircraft ABU-8/A
or
Accelerometer, Aircraft ABU-8A/A.

5.2.1 Reinspection date. The reinspection date markings shall be as specified by the procuring activity.

6. NOTES

*6.1 Intended use. The ABU-8/A and ABU-8A/A accelerometers covered by this specification are intended for use in aircraft to indicate vertical acceleration during maneuvers and in rough air.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type of accelerometer required (see 1.2)

c. When sampling plan B and C tests will not be conducted

d. Levels of packaging and packing and reinspection date markings (see section 5).

*6.2.1 Modification of equipment. If the accelerometers need to be modified due to failures during reliability testing (see 4.6.21), the contractor will accomplish the modification to the approved configuration at no increase in cost to the Government.

*6.3 Test report. The contractor will submit three copies of the test report specified in 4.4.2 to the procuring activity.

6.4 Definitions. For the purpose of this specification, the following definitions will apply:

a. Tapped lightly: The application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments

b. Hermetic seal: A perfectly closed and airtight seal made between vitric and metallic materials or between vitric and vitric and metallic and metallic materials. This does not include seals accomplished by gaskets

c. G-unit: An acceleration of 32.2 feet per second per second

d. Amplitude: The extent of motion as measured from one extreme to the opposite extreme.

*6.5 International standardization. Certain provisions of 3.4.1, 3.7, 3.9, and 3.10 of this specification are the subject of international standardization agreements ABC 10/4A and STANAG 3330. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.6 The margins of the specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activities:
Air Force - 11, 71

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:25Z
Check the source to verify that this is the current version before use.